

Lab 02

Git and GitHub – Branching, Merging, and Pull Request Workflow

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BCS-7A

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GitHub Username: sidraliaqat

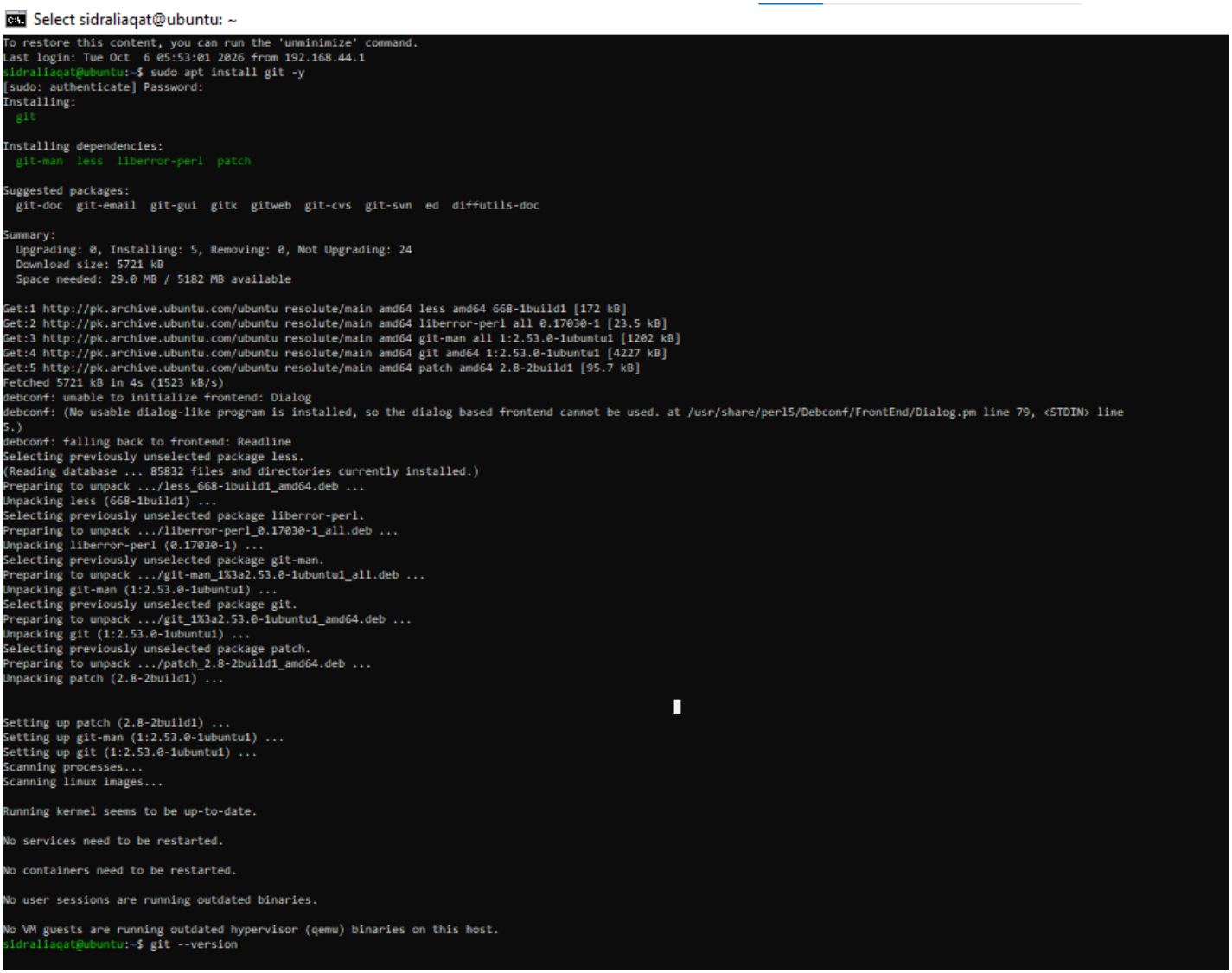
Getting Started and Tasks 1 to 2: Setup

All commands were run on the Ubuntu VM, where the prompt reads `sidraliaqat@ubuntu`.

Install Git

```
sudo apt install git -y
git --version
```

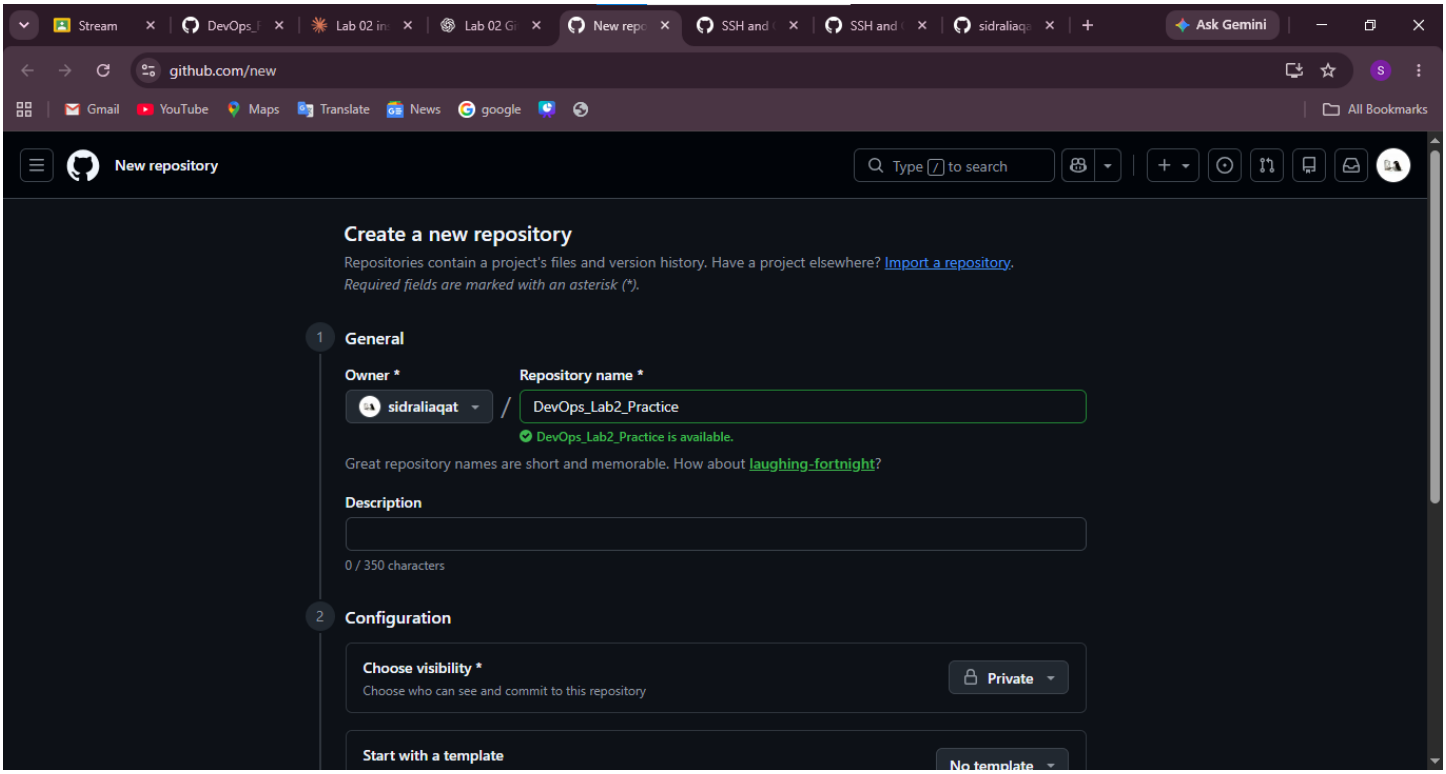
Screenshot: git_installation.png



Task 1: Create a private GitHub repository

Created a private repository named `DevOps_Lab2_Practice` on GitHub.

Screenshot: repo_private.png



Task 2: Connect the repository via SSH

Generated an ed25519 key and displayed the public key.

```
ssh-keygen -t ed25519 -C "sidraliaqat204@gmail.com"
cat ~/.ssh/id_ed25519.pub
```

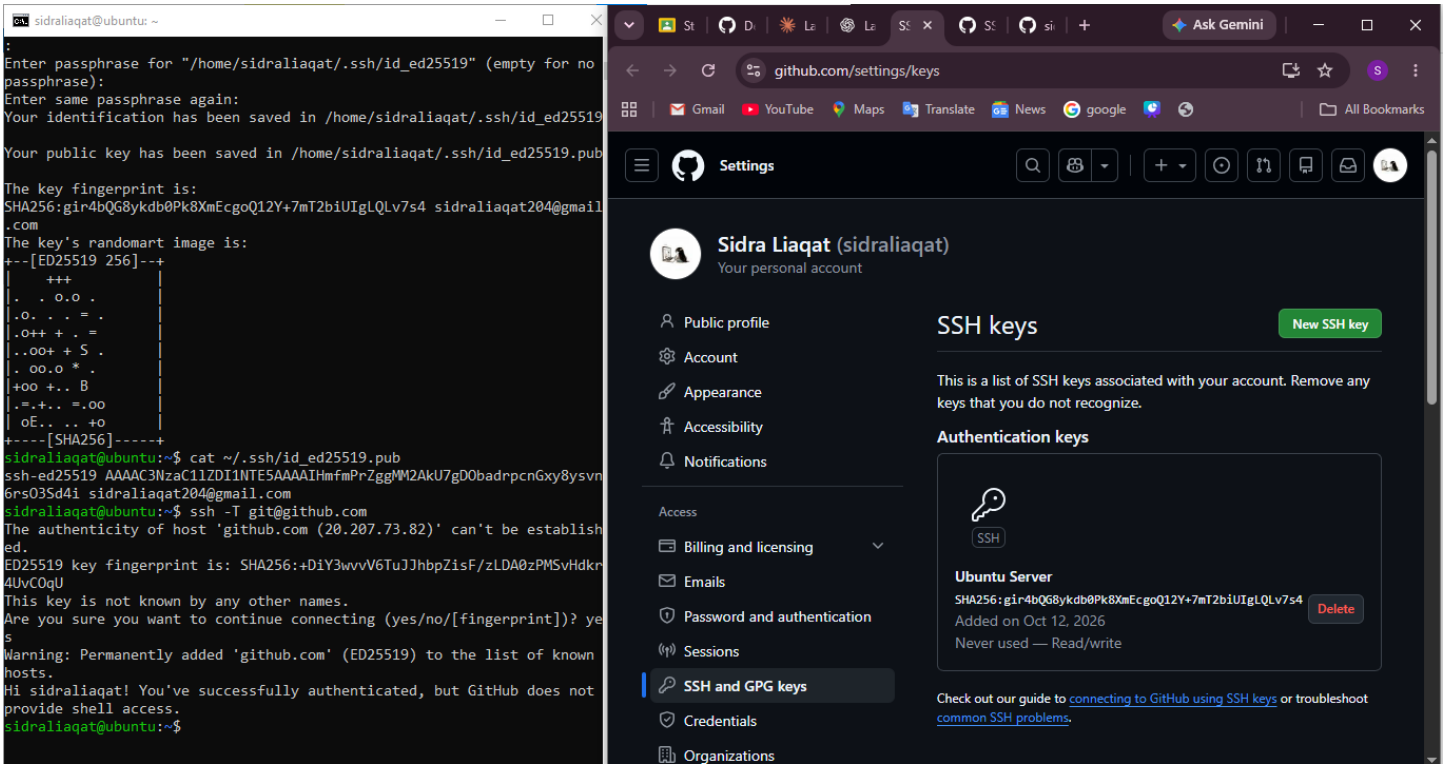
Screenshot: ssh_keygen.png

```
git version 2.53.0
sidraliaqat@ubuntu:~$ ssh-keygen -t ed25519 -C "sidraliaqat204@gmail.com"
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/sidraliaqat/.ssh/id_ed25519):
Enter passphrase for "/home/sidraliaqat/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/sidraliaqat/.ssh/id_ed25519
Your public key has been saved in /home/sidraliaqat/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:gir4bQG8ykdb0Pk8XmEcgoQ12Y+7mT2biUIgLQLv7s4 sidraliaqat204@gmail.com
The key's randomart image is:
+--[ED25519 256]--+
|    +++          |
|.  . o.o .       |
|.o. . . = .      |
|.o++ + . =       |
|..oo+ + S .      |
|. oo.o * .       |
|+oo +.. B        |
|.=.+.. =.oo      |
| oE.. .. +o      |
+----[SHA256]-----+
sidraliaqat@ubuntu:~$
```

Added the public key to GitHub under Settings > SSH and GPG keys and tested it.

```
ssh -T git@github.com
```

Screenshot: github_sshkey.png



Cloned the repository over SSH.

```
git clone git@github.com:sidraliaqat/DevOps_Lab2_Practice.git
```

Screenshot: ssh_clone.png

```
sidraliaqat@ubuntu:~$ git clone git@github.com:sidraliaqat/DevOps_Lab2_Practice.git
Cloning into 'DevOps_Lab2_Practice'...
warning: You appear to have cloned an empty repository.
```

Tasks 3 to 6: Identity, .git Folder, Local Repo, Staging

Task 3: Configure Git username and email

```
git config --global user.name "Sidra Liaqat"
git config --global user.email "sidraliaqat204@gmail.com"
```

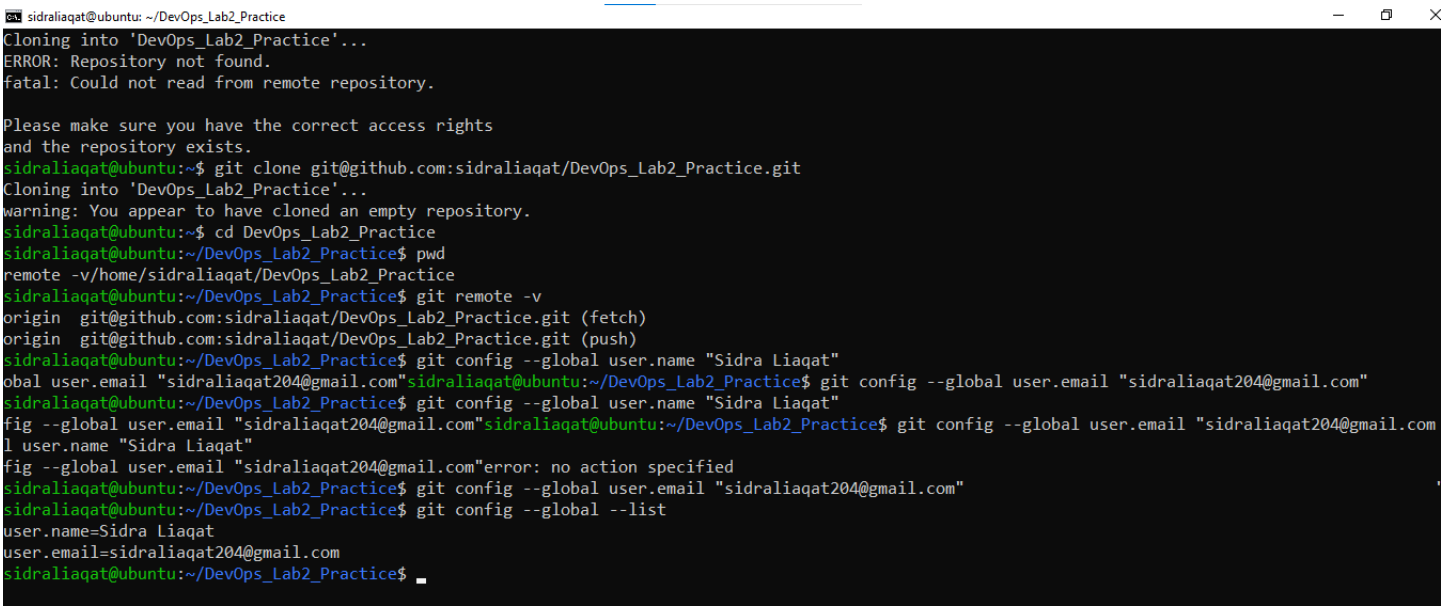
Screenshot: git_identity.png

```
sidraliaqat@ubuntu:~$ git clone git@github.com:sidraliaqat/DevOps_Lab2_Practice.git
Cloning into 'DevOps_Lab2_Practice'...
warning: You appear to have cloned an empty repository.
sidraliaqat@ubuntu:~$ cd DevOps_Lab2_Practice
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ pwd
remote -v/home/sidraliaqat/DevOps_Lab2_Practice
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git remote -v
origin  git@github.com:sidraliaqat/DevOps_Lab2_Practice.git (fetch)
origin  git@github.com:sidraliaqat/DevOps_Lab2_Practice.git (push)
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git config --global user.name "Sidra Liaqat"
obal user.email "sidraliaqat204@gmail.com"sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git config --global user.email "sidraliaqat204@gmail.com"
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git config --global user.name "Sidra Liaqat"
fig --global user.email "sidraliaqat204@gmail.com"sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git config --global user.email "sidraliaqat204@gmail.com"
l user.name "Sidra Liaqat"
fig --global user.email "sidraliaqat204@gmail.com"error: no action specified
```

Verified the configuration.

```
git config --list
```

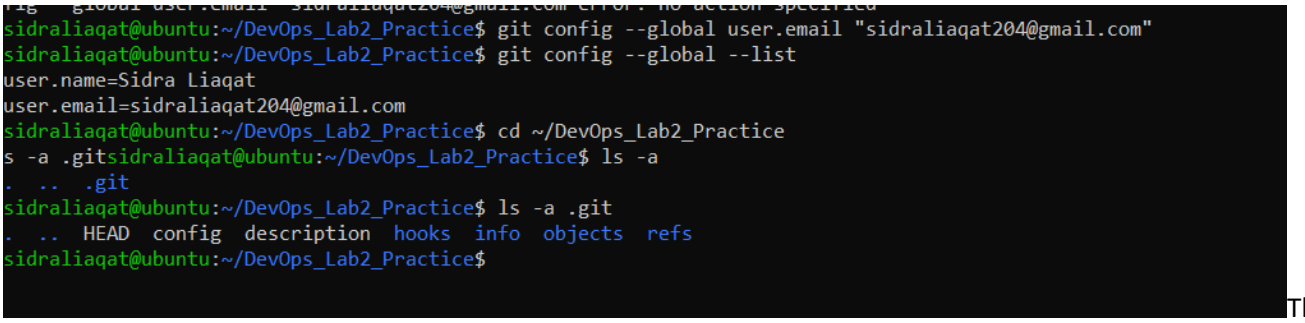
Screenshot: git_config_list.png



Task 4: Explore the .git folder

```
cd DevOps_Lab2_Practice
ls -a .git
```

Screenshot: git_folder.png



The `.git` folder holds the repository's history, objects, refs, logs and config.

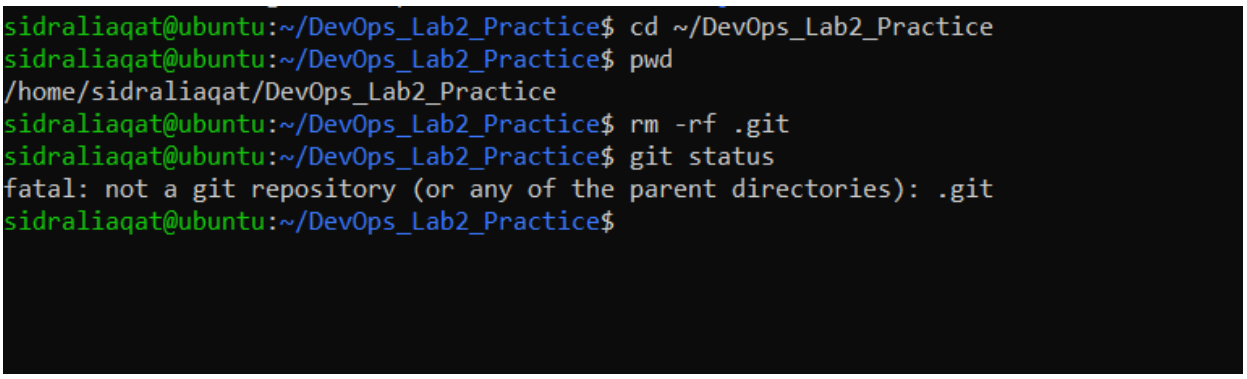
Task 5: Local repository management

Deleted the `.git` folder and ran `git status`.

```
rm -rf .git
git status
```

`git status` fails with `fatal: not a git repository` because the `.git` folder is what makes a directory a Git repository. Without it, Git has no history or tracking data to report on.

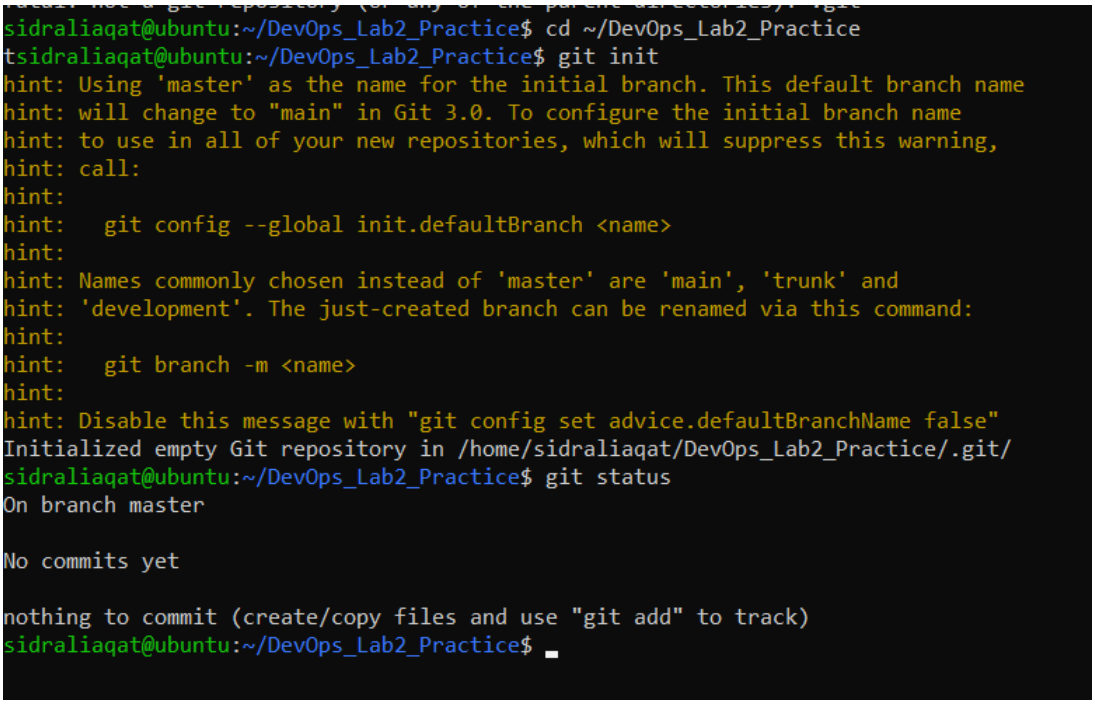
Screenshot: delete_git.png



Re-initialized the repository.

```
git init
```

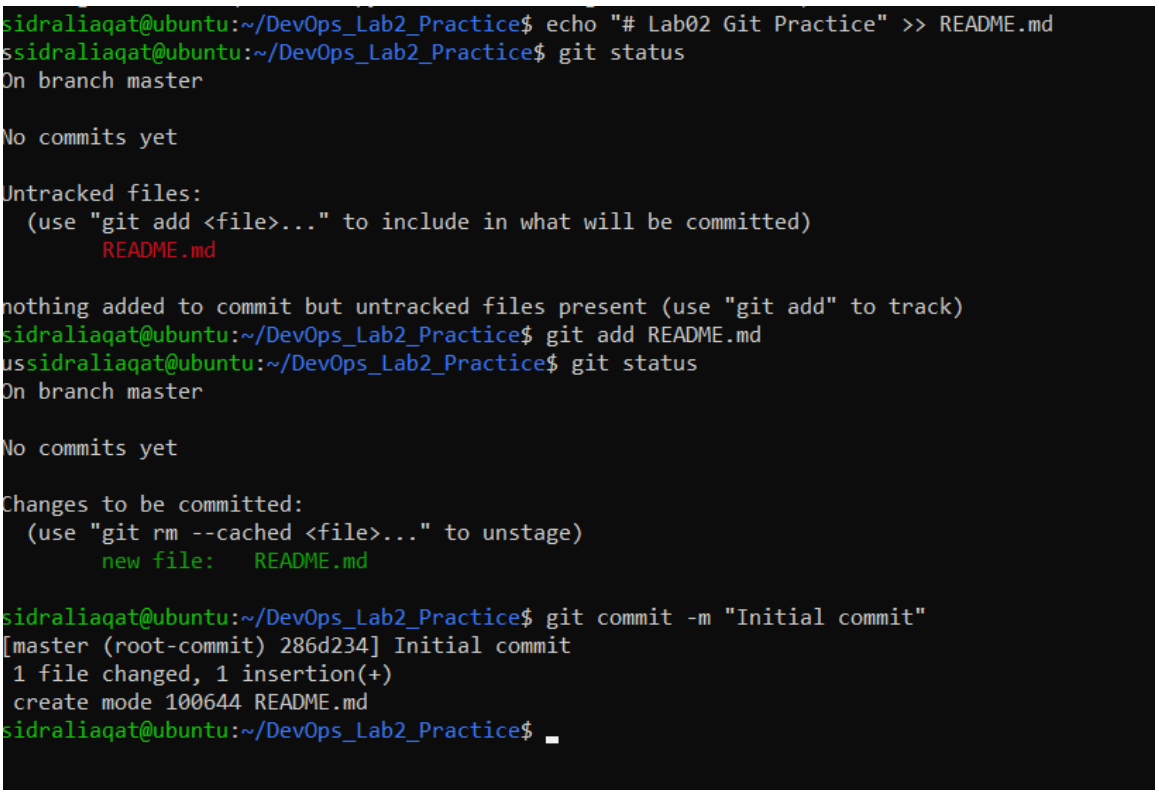
Screenshot: git_init.png



Created `README.md` and made the first commit.

```
echo "# Lab02 Git Practice" >> README.md
git add README.md
git commit -m "Initial commit"
```

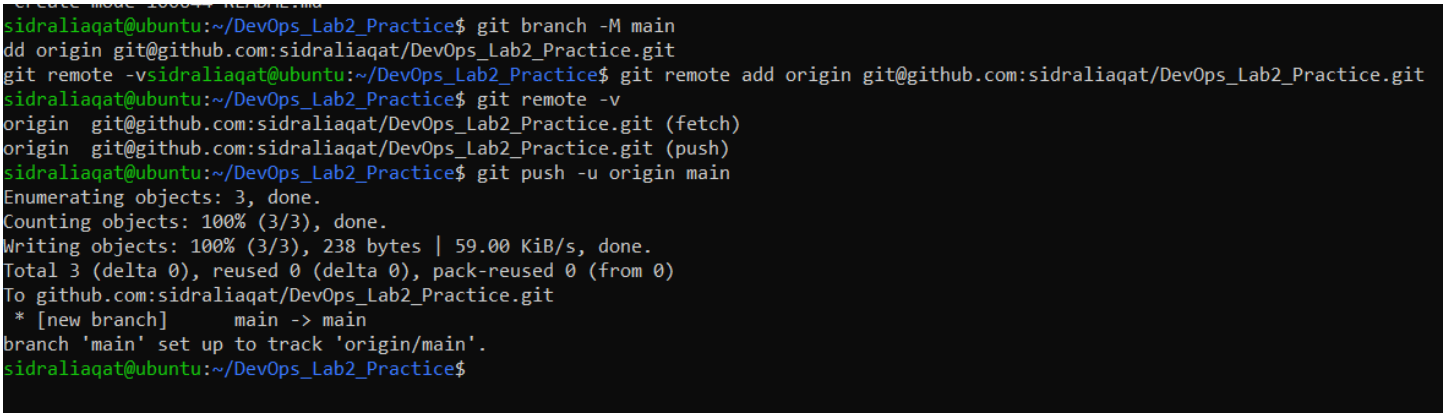
Screenshot: first_commit.png



Connected the local repository to GitHub and pushed.

```
git remote add origin git@github.com:sidraliaqat/DevOps_Lab2_Practice.git
git branch -M main
git push -u origin main
```

Screenshot: first_push.png

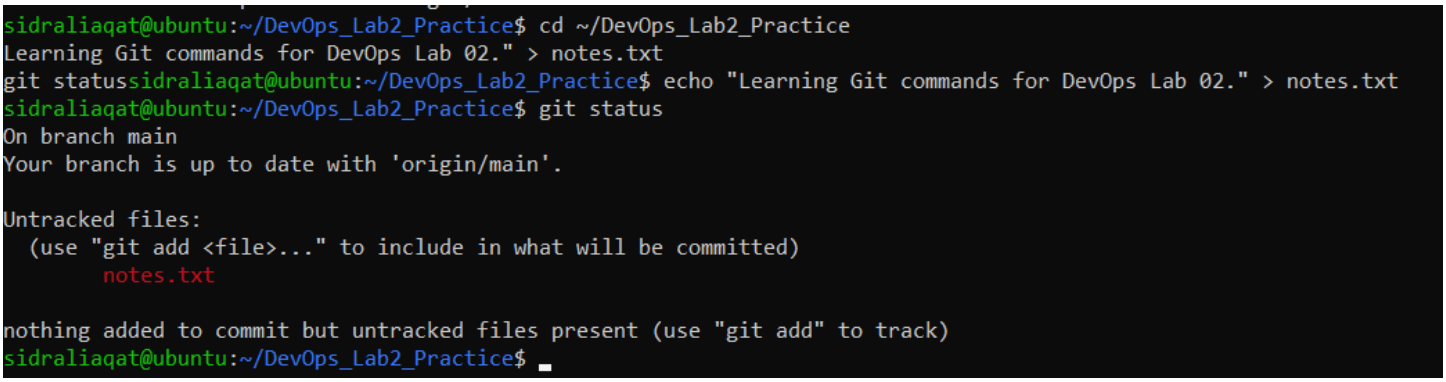


Task 6: File status and staging

Created `notes.txt` and checked the status.

```
echo "Adding some test notes" >> notes.txt
git status
```

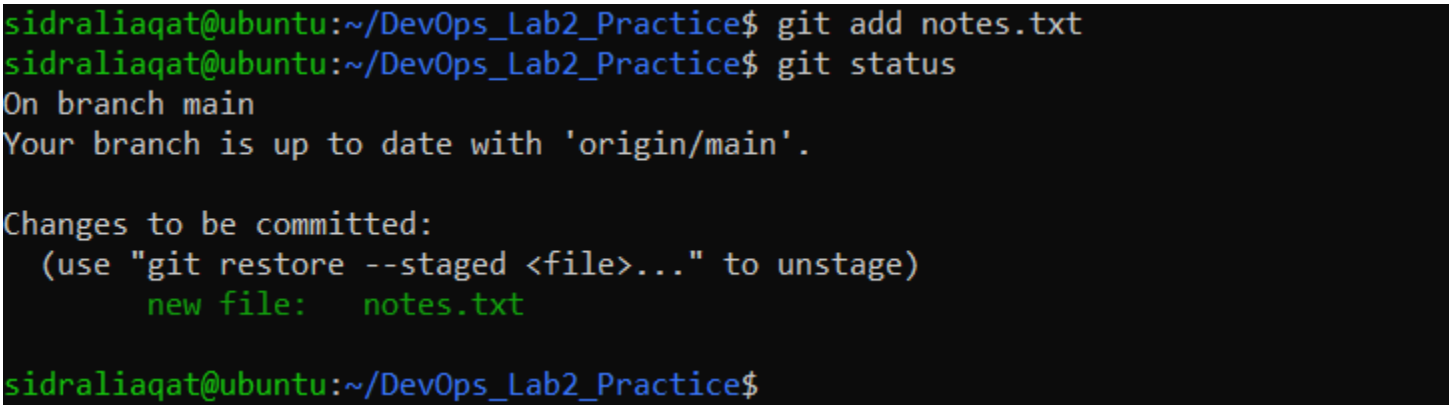
Screenshot: status1.png



Staged and committed the file.

```
git add notes.txt
git commit -m "Add notes.txt"
```

Screenshot: commit_notes.png

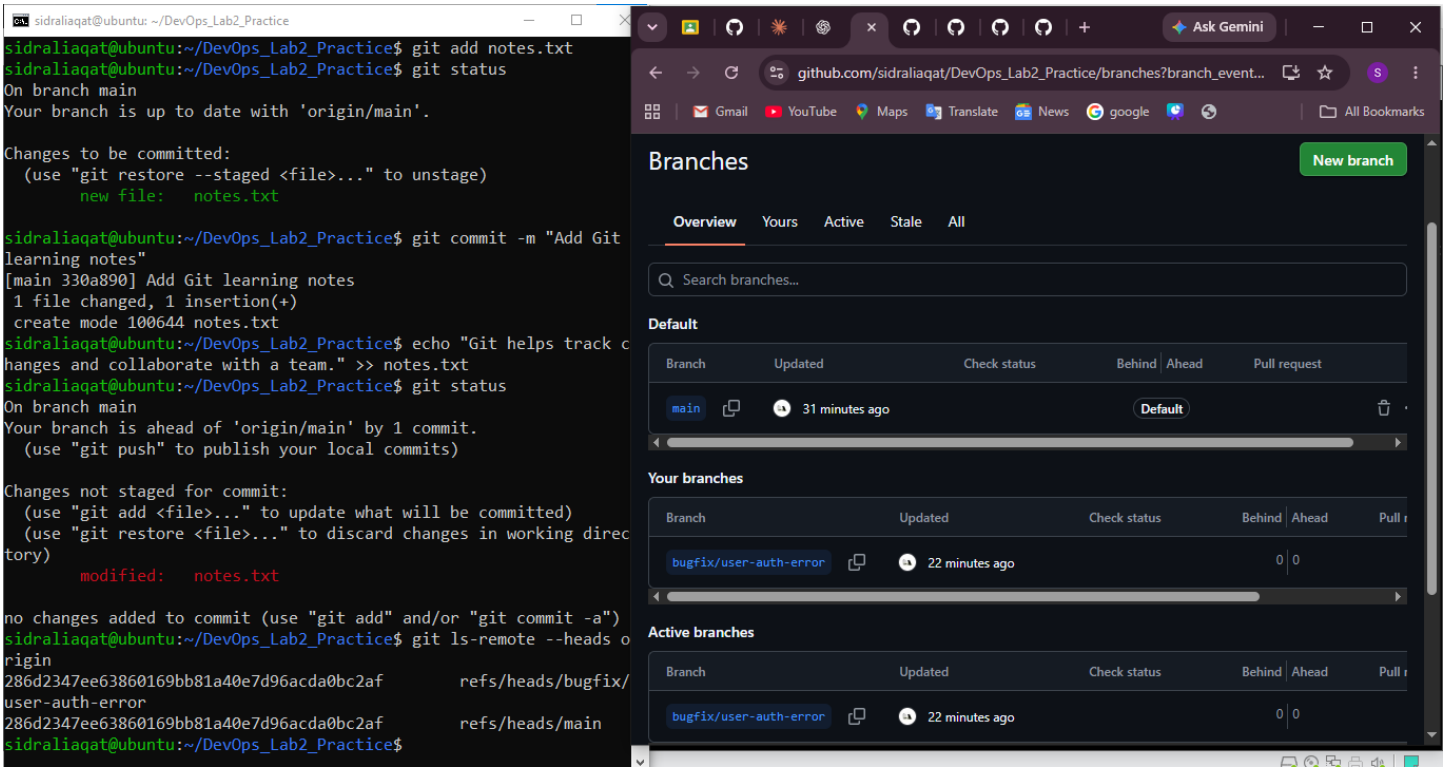


Tasks 7 to 9: Branching and Merging

Task 7: Branch creation using the GitHub GUI

Created `bugfix/user-auth-error` on GitHub.

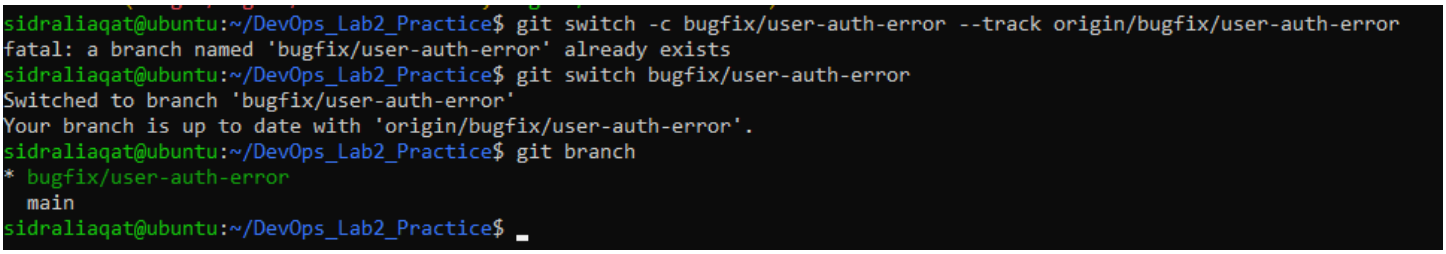
Screenshot: bugfix_branch_gui.png



Pulled the branch into the local repository and listed the branches.

```
git pull origin bugfix/user-auth-error
git branch -a
```

Screenshot: bugfix_branch_local.png



Task 8: Branch creation and push using Git Bash

```
git checkout -b feature/db-connection
git push origin feature/db-connection
```

Screenshot: feature_db_branch.png

```
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git branch
* bugfix/user-auth-error
  main
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ it status
-bash: it: command not found
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git status
On branch bugfix/user-auth-error
Your branch is up to date with 'origin/bugfix/user-auth-error'.

nothing to commit, working tree clean
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git switch main
 origin main
git switch -c feature/db-connectionSwitched to branch 'main'
Your branch is ahead of 'origin/main' by 2 commits.
  (use "git push" to publish your local commits)
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git pull origin main

From github.com:sidraliaqat/DevOps_Lab2_Practice
 * branch            main       -> FETCH_HEAD
Already up to date.
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git switch -c feature/db-connection
Switched to a new branch 'feature/db-connection'
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git branch
  bugfix/user-auth-error
* feature/db-connection
  main
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$
```

Task 9: Branching and merging

Created and switched to `feature-1`.

```
git checkout -b feature-1
```

Screenshot: branch_create.png

sidraliaqat@ubuntu: ~/DevOps_Lab2_Practice

```
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git checkout main
Already on 'main'
Your branch is behind 'origin/main' by 2 commits, and can be fast-forwarded.
  (use "git pull" to update your local branch)
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git checkout -b feature-1
Switched to a new branch 'feature-1'
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$
```

Added a function to `main.py` and committed it.

```
git add main.py
git commit -m "Add new function to main.py"
```

Screenshot: feature_commit.png

```
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ cat > main.py <<'EOF'
connect_to_dat> def connect_to_database():
>     print("Database connection configured.")
>
> if __name__ == "__main__":
>     connect_to_database()
> EOF
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ cat main.py
def connect_to_database():
    print("Database connection configured.")

if __name__ == "__main__":
    connect_to_database()
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git status
On branch feature/db-connection
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        main.py

nothing added to commit but untracked files present (use "git add" to track)
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git add main.py
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git status
On branch feature/db-connection
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   main.py

sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Add database connection function"
[feature/db-connection a5085fe] Add database connection function
 1 file changed, 5 insertions(+)
 create mode 100644 main.py
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$
```

Switched back to `main` and merged `feature-1`.

```
git checkout main
git merge feature-1
```

Screenshot: merge.png

```
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git checkout main
rge feature-1Switched to branch 'main'
Your branch is behind 'origin/main' by 2 commits, and can be fast-forwarded.
  (use "git pull" to update your local branch)
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git merge feature-1
Already up to date.
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$
```

Merging straight into `main` from a local terminal is risky on a team, because nobody reviews the change first. A Pull Request lets others review it before it reaches `main`, which is why Task 10 uses one.

Pushed both branches.

```
git push origin main
git push origin feature-1
```

Screenshot: push_branches.png

sidraliaqat@ubuntu: ~/DevOps_Lab2_Practice

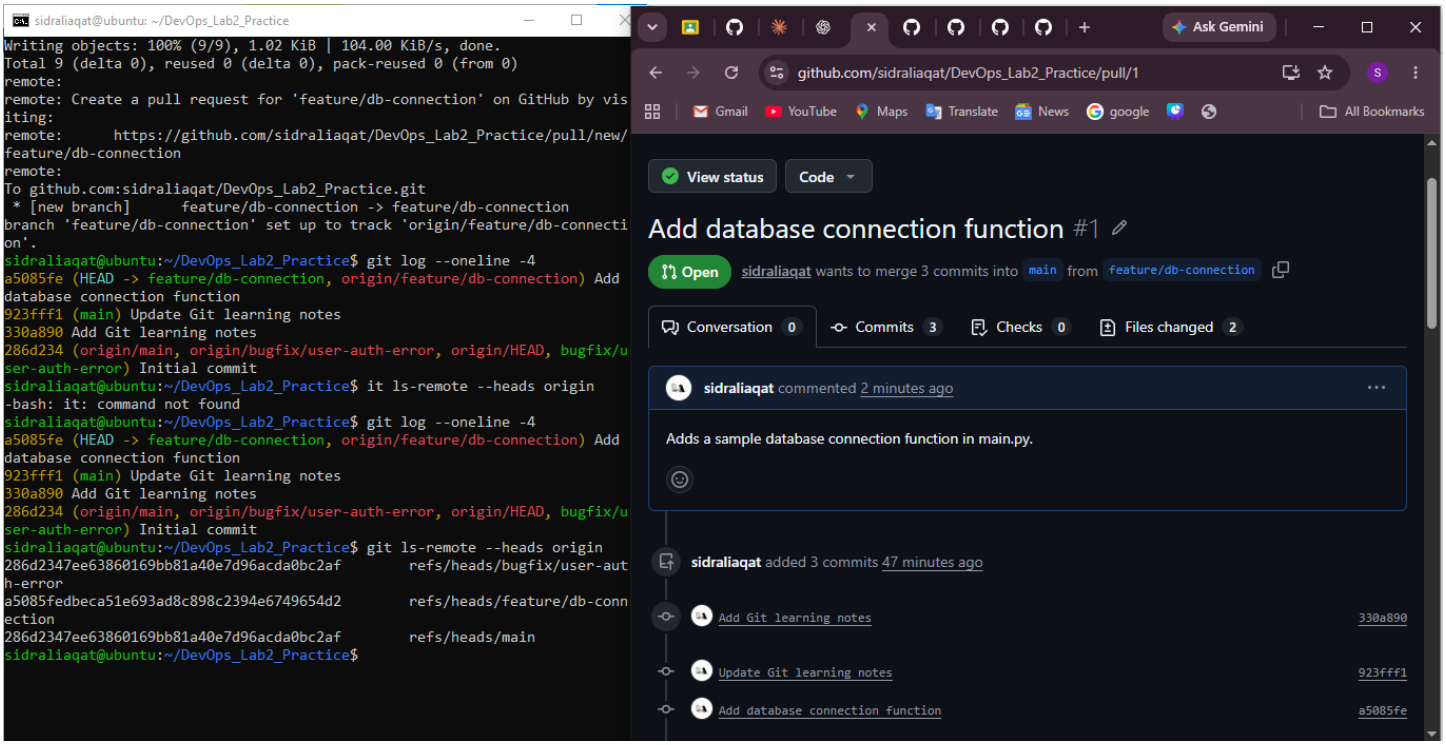
```
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git branch
  bugfix/user-auth-error
* feature/db-connection
  main
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git push -u origin feature/db-connection
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 2 threads
Compressing objects: 100% (8/8), done.
Writing objects: 100% (9/9), 1.02 KiB | 104.00 KiB/s, done.
Total 9 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature/db-connection' on GitHub by visiting:
remote:      https://github.com/sidraliaqat/DevOps_Lab2_Practice/pull/new/feature/db-connection
remote:
To github.com:sidraliaqat/DevOps_Lab2_Practice.git
 * [new branch]      feature/db-connection -> feature/db-connection
branch 'feature/db-connection' set up to track 'origin/feature/db-connection'.
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ _
```

Tasks 10 and 11: Pull Requests and Branch Strategy

Task 10: Pull request and branch review (GitHub GUI)

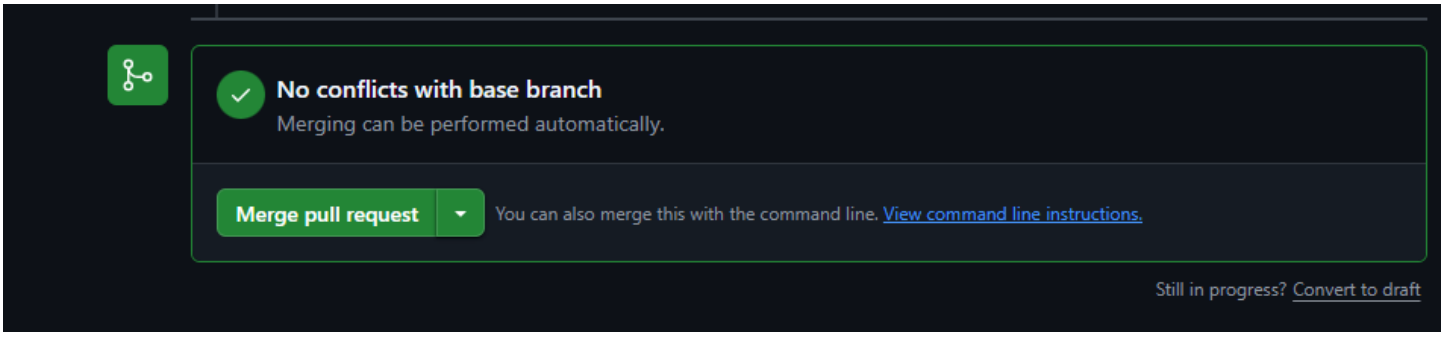
Opened PR #1, "Add database connection function", from `feature/db-connection` into `main`.

Screenshot: pr_creation.png



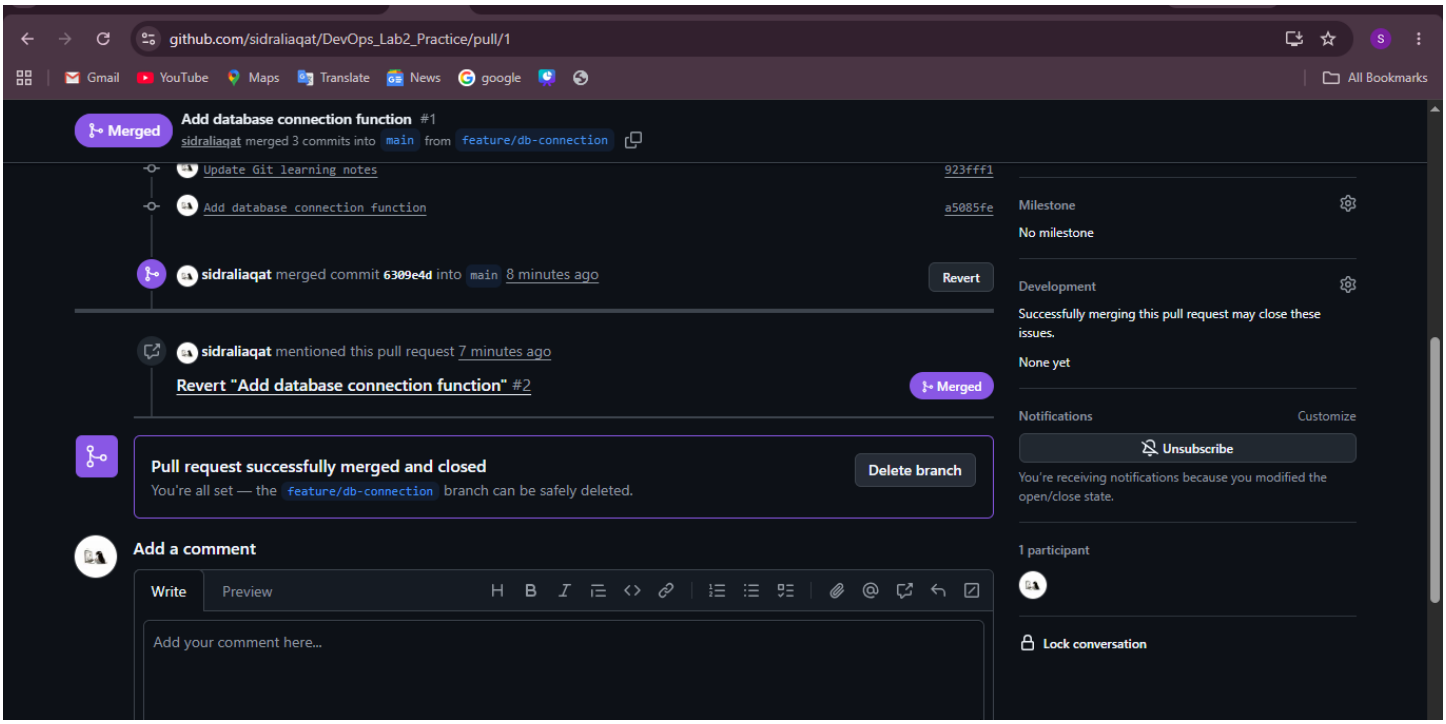
Reviewed the PR and merged it with the GitHub GUI.

Screenshot: pr_merge.png



Deleted the `feature/db-connection` branch after the merge.

Screenshot: branch_delete.png



Task 11: Detailed branch strategy (develop and staging)

Workflow: developers work on `feature/*` and `bugfix/*` branches, which merge into `develop`. `develop` merges into `staging` for testing, and `staging` merges into `main` (production).

Created the `develop` and `staging` branches and listed all branches.

```
git checkout main
git checkout -b develop
git push origin develop
git checkout -b staging
git push origin staging
git branch -a
```

Screenshot: branch_strategy.png

```
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git branch -a
  bugfix/user-auth-error
* main
  remotes/origin/HEAD -> origin/main
  remotes/origin/bugfix/user-auth-error
  remotes/origin/main
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$
```

Merged a feature branch into `develop`, then `develop` into `staging`.

```
git checkout develop
git merge feature-1
git checkout staging
git merge develop
```

Screenshot: branch_merges.png

```
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git log --oneline --graph --all --decorate
*   94b339f (HEAD -> main, origin/main, origin/HEAD) Merge pull request #2 from sidraliaqat/revert-1-feature/db-connection
|\
| * 873fcd9 Revert "Add database connection function"
|/
*   6309e4d Merge pull request #1 from sidraliaqat/feature/db-connection
|\
| * a5085fe Add database connection function
| * 923fff1 Update Git learning notes
| * 330a890 Add Git learning notes
|/
* 286d234 (origin/bugfix/user-auth-error, bugfix/user-auth-error) Initial commit
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$
```

Merged `staging` into `main` and pushed.

```
git checkout main
git merge staging
git push origin main
```

Screenshot: final_merge.png

```
sidraliaqat@ubuntu: ~/DevOps_Lab2_Practice
  bugfix/user-auth-error
* main
  remotes/origin/HEAD -> origin/main
  remotes/origin/bugfix/user-auth-error
  remotes/origin/main
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git log --oneline --graph --all --decorate
*   94b339f (HEAD -> main, origin/main, origin/HEAD) Merge pull request #2 from sidraliaqat/revert-1-feature/db-connection
|\
| * 873fcd9 Revert "Add database connection function"
|/
*   6309e4d Merge pull request #1 from sidraliaqat/feature/db-connection
|\
| * a5085fe Add database connection function
| * 923fff1 Update Git learning notes
| * 330a890 Add Git learning notes
|/
* 286d234 (origin/bugfix/user-auth-error, bugfix/user-auth-error) Initial commit
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git switch main
 log --oneline --graph --decorate -10Already on 'main'
Your branch is up to date with 'origin/main'.
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git pull origin main

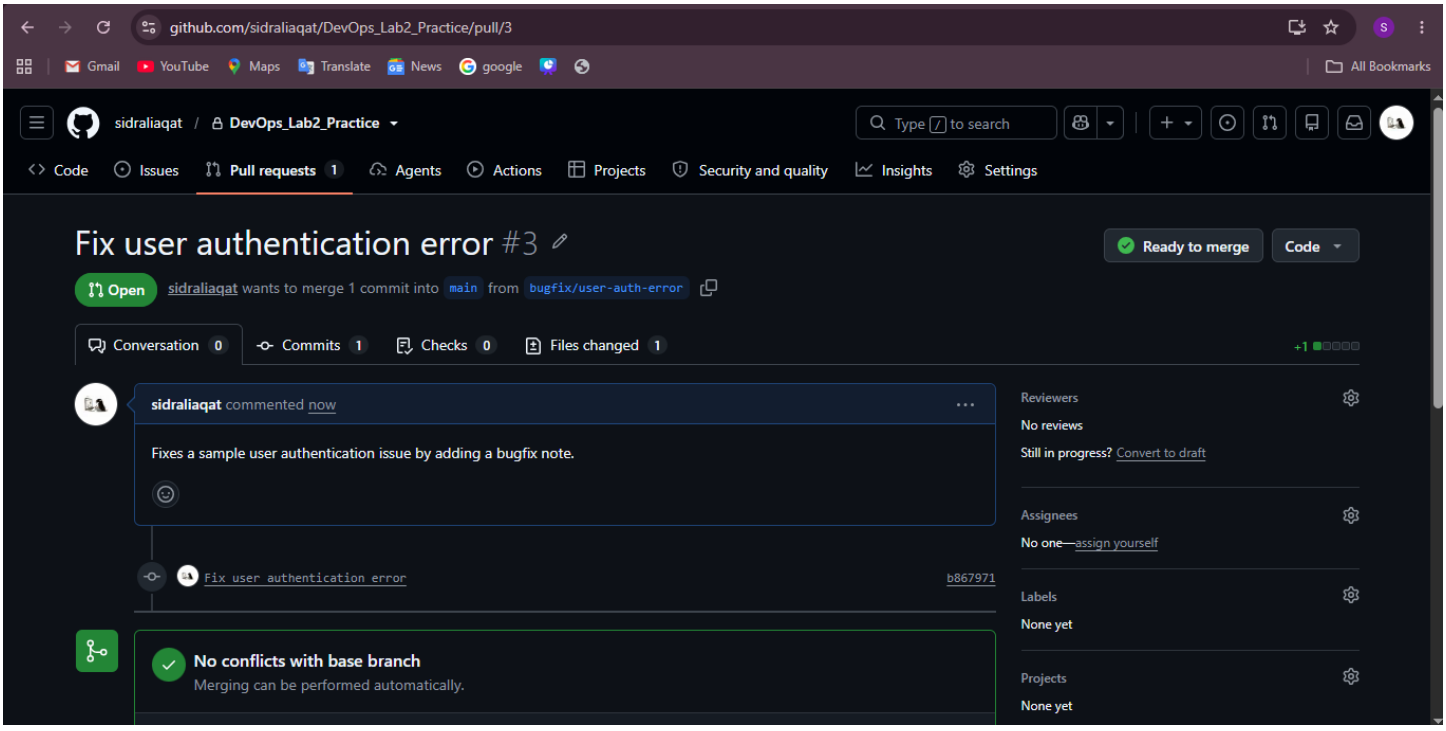
From github.com:sidraliaqat/DevOps_Lab2_Practice
 * branch            main       -> FETCH_HEAD
Already up to date.
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git log --oneline --graph --decorate -10
*   94b339f (HEAD -> main, origin/main, origin/HEAD) Merge pull request #2 from sidraliaqat/revert-1-feature/db-connection
|\
| * 873fcd9 Revert "Add database connection function"
|/
*   6309e4d Merge pull request #1 from sidraliaqat/feature/db-connection
|\
| * a5085fe Add database connection function
| * 923fff1 Update Git learning notes
| * 330a890 Add Git learning notes
|/
* 286d234 (origin/bugfix/user-auth-error, bugfix/user-auth-error) Initial commit
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$
```

Tasks 12 and 13: Code Review and Branch Cleanup

Task 12: Code review workflow

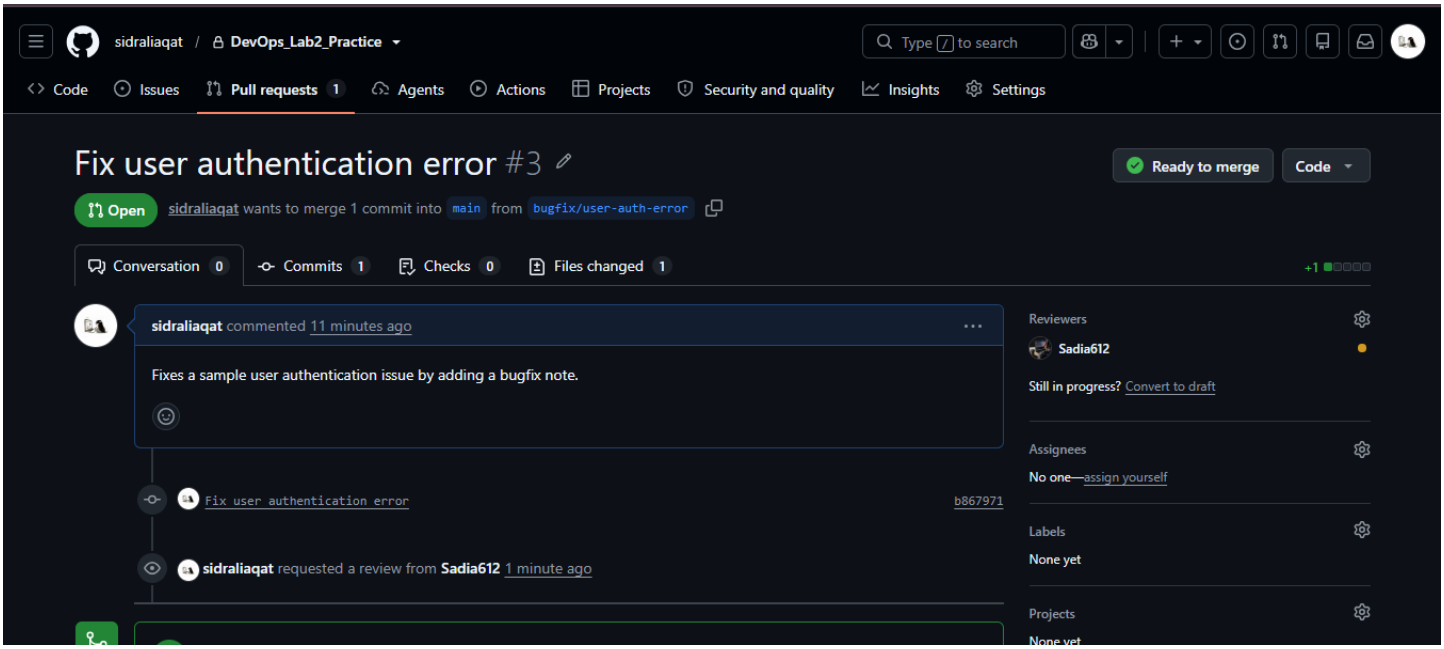
Created a PR from a feature branch into `main` with a clear title and description.

Screenshot: pr_create_details.png



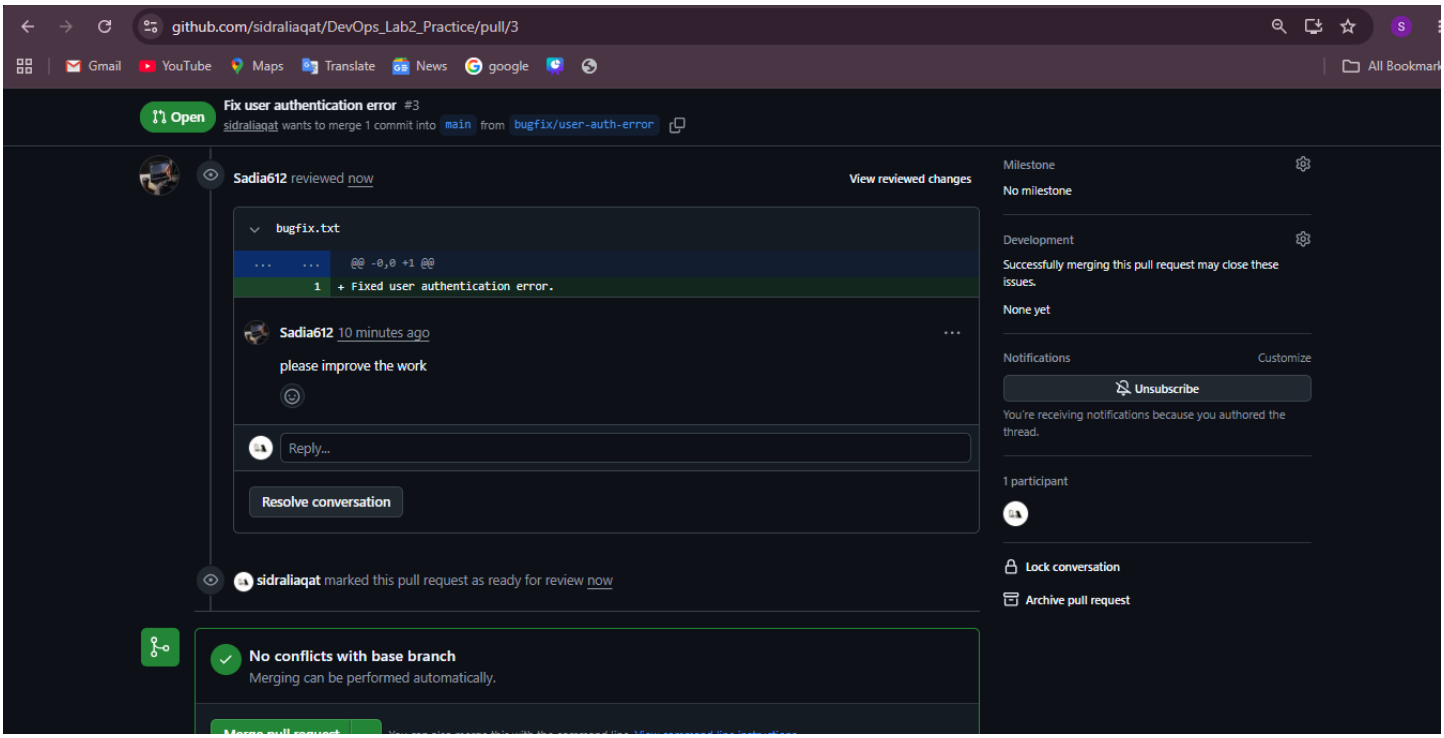
Assigned a reviewer.

Screenshot: pr_assigned_reviewer.png



Reviewer requested changes and left comments.

Screenshot: pr_request_changes.png



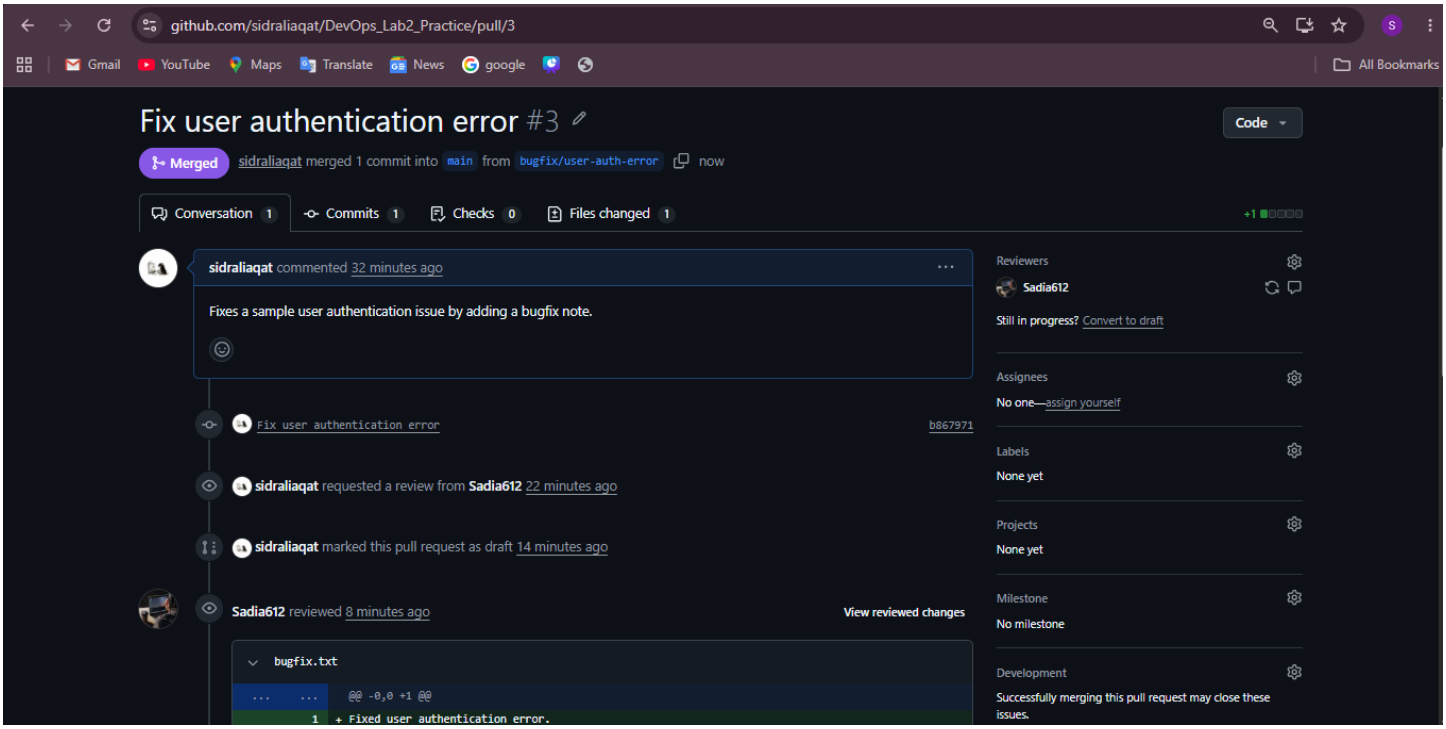
Closed a separate practice PR without merging it.

Screenshot: pr_updated_with_commits.png

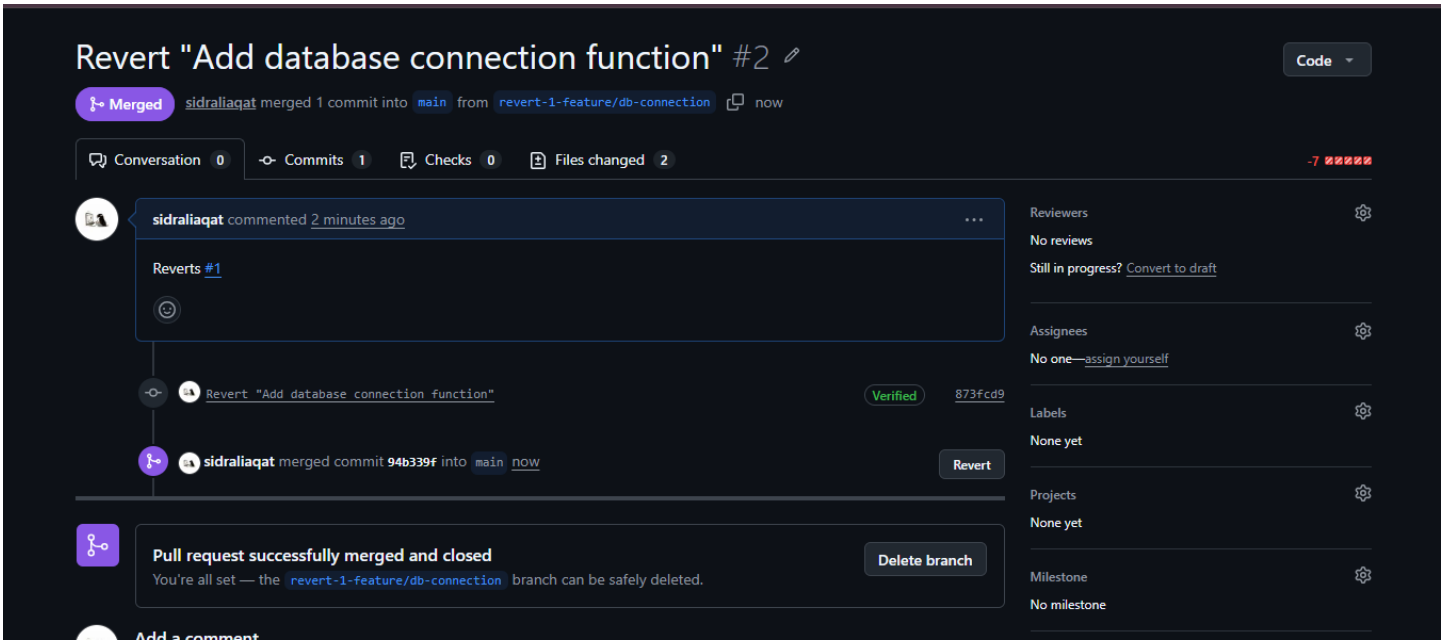
```
sidraliaqat@ubuntu: ~/DevOps_Lab2_Practice
| * 873fcd9 Revert "Add database connection function"
|/
*   6309e4d Merge pull request #1 from sidraliaqat/feature/db-connection
|\
| * a5085fe Add database connection function
| * 923fff1 Update Git learning notes
| * 330a890 Add Git learning notes
|/
* 286d234 (origin/bugfix/user-auth-error, bugfix/user-auth-error) Initial commit
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git switch -c bugfix/user-auth-error --track origin/bugfix/user-auth-error
fatal: a branch named 'bugfix/user-auth-error' already exists
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git switch bugfix/user-auth-error
Switched to branch 'bugfix/user-auth-error'
Your branch is up to date with 'origin/bugfix/user-auth-error'.
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git branch
* bugfix/user-auth-error
  main
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git switch bugfix/user-auth-error
Already on 'bugfix/user-auth-error'
Your branch is up to date with 'origin/bugfix/user-auth-error'.
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ echo "Fixed user authentication error." > bugfix.txt
 bugfix.txt
git commit -m "Fix user authentication error"sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git add bugfix.txt
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Fix user authentication error"
[bugfix/user-auth-error b867971] Fix user authentication error
 1 file changed, 1 insertion(+)
 create mode 100644 bugfix.txt
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git push origin bugfix/user-auth-error
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 328 bytes | 164.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:sidraliaqat/DevOps_Lab2_Practice.git
   286d234..b867971  bugfix/user-auth-error -> bugfix/user-auth-error
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git switch bugfix/user-auth-error
Already on 'bugfix/user-auth-error'
Your branch is up to date with 'origin/bugfix/user-auth-error'.
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ echo "This fix documents the user authentication issue and its resolution." >> bugfix.txt
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$
```

Merged the PR after approval. Merge style used: merge commit.

Screenshot: pr_merge_confirm.png

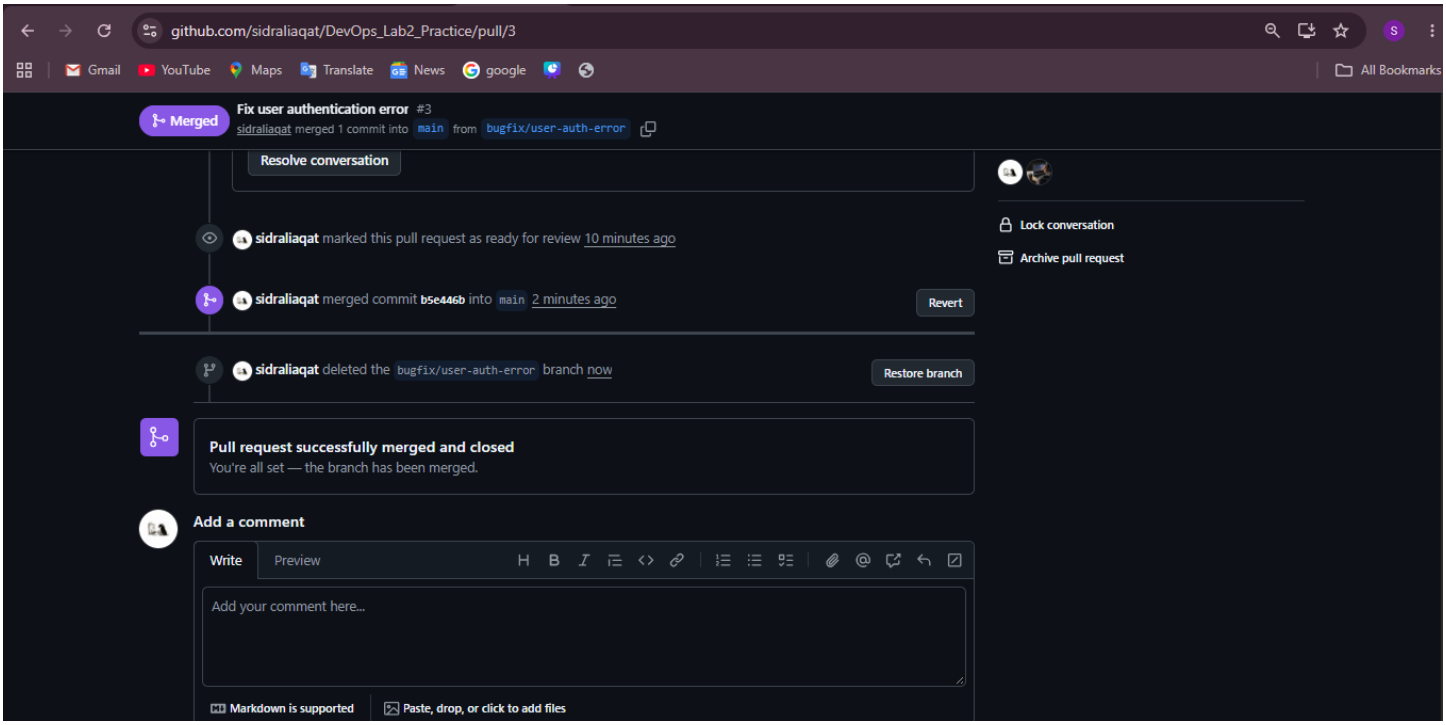


Screenshot: pr_merged.png



Deleted the merged PR branch.

Screenshot: pr_branch_deleted.png



Task 13: Branch cleanup best practices

Deleted a merged branch through the GitHub UI.

Screenshot: remote_branch_deleted.png

```
Select sidraliaqat@ubuntu: ~/DevOps_Lab2_Practice
Please commit your changes or stash them before you switch branches.
Aborting
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git pull origin main

remote: Enumerating objects: 1, done.
remote: Counting objects: 100% (1/1), done.
remote: Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (1/1), 918 bytes | 459.00 KiB/s, done.
From github.com:sidraliaqat/DevOps_Lab2_Practice
 * branch            main       -> FETCH_HEAD
   94b339f..b5e446b  main       -> origin/main
Updating b867971..b5e446b
Fast-forward
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git fetch --prune
From github.com:sidraliaqat/DevOps_Lab2_Practice
 - [deleted]         (none)     -> origin/bugfix/user-auth-error
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git branch -a
* bugfix/user-auth-error
  main
  remotes/origin/HEAD -> origin/main
  remotes/origin/main
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git add bugfix.txt
 "Improve user authentication fix notes"sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git commit -m "Improve user authentication fix notes"
[bugfix/user-auth-error 046a152] Improve user authentication fix notes
 1 file changed, 1 insertion(+)
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git switch main
gfix/user-auth-error
git branch -aSwitched to branch 'main'
Your branch is behind 'origin/main' by 2 commits, and can be fast-forwarded.
  (use "git pull" to update your local branch)
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git branch -d bugfix/user-auth-error
error: the branch 'bugfix/user-auth-error' is not fully merged
hint: If you are sure you want to delete it, run 'git branch -D bugfix/user-auth-error'
hint: Disable this message with "git config set advice.forceDeleteBranch false"
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git branch -a
  bugfix/user-auth-error
* main
  remotes/origin/HEAD -> origin/main
  remotes/origin/main
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$
```

Deleted another completed remote branch from the terminal.

```
git push origin --delete <branch-name>
```

Screenshot: remote_branch_delete_cmd.png

```
sidraliaqat@ubuntu: ~/DevOps_Lab2_Practice
 * branch            main       -> FETCH_HEAD
Already up to date.
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git branch
  bugfix/user-auth-error
  feature/db-connection
* main
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git branch -d feature/db-connection
Deleted branch feature/db-connection (was a5085fe).
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git switch main
 origin main
git branchAlready on 'main'
Your branch is up to date with 'origin/main'.
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git pull origin main


From github.com:sidraliaqat/DevOps_Lab2_Practice
 * branch            main       -> FETCH_HEAD
Already up to date.
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git branch
  bugfix/user-auth-error
* main
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git branch -d feature/db-connection
error: branch 'feature/db-connection' not found
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git branch -a
  bugfix/user-auth-error
* main
  remotes/origin/HEAD -> origin/main
  remotes/origin/bugfix/user-auth-error
  remotes/origin/feature/db-connection
  remotes/origin/main
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git push origin --delete feature/db-connection
error: unable to delete 'feature/db-connection': remote ref does not existerror: failed to push some refs to 'github.com:sidraliaqat/DevOps_Lab2_Practice.git'
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git fetch --prune

From github.com:sidraliaqat/DevOps_Lab2_Practice
 - [deleted]         (none)     -> origin/feature/db-connection
sidraliaqat@ubuntu:~/DevOps_Lab2_Practice$ git branch -a
  bugfix/user-auth-error
* main
  remotes/origin/HEAD -> origin/main
  remotes/origin/bugfix/user-auth-error
```

Updated the local repository.

```
git checkout main
git pull origin main
```

```
git branch -d <branch-name>
git branch
```
Deleting merged branches keeps the repository tidy and avoids confusion about which branches are still active.